

Thermodynamics

Instructor

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Chapter 13 – Problems

**Problem numbers correspond to
5th Edition**

13-13 A gas mixture consists of 5 kg of O_2 , 8 kg of N_2 , and 10 kg of CO_2 . Determine (a) the mass fraction of each component, (b) the mole fraction of each component, and (c) the average molar mass and gas constant of the mixture.

Properties The molar masses of O_2 , N_2 , and CO_2 are 32.0, 28.0 and 44.0 kg/kmol, respectively (Table A-1)

Analysis (a) The total mass of the mixture is

$$m_m = m_{O_2} + m_{N_2} + m_{CO_2} = 5 \text{ kg} + 8 \text{ kg} + 10 \text{ kg} = 23 \text{ kg}$$

Then the mass fraction of each component becomes

$$mf_{O_2} = \frac{m_{O_2}}{m_m} = \frac{5 \text{ kg}}{23 \text{ kg}} = 0.217$$

$$mf_{N_2} = \frac{m_{N_2}}{m_m} = \frac{8 \text{ kg}}{23 \text{ kg}} = 0.348$$

$$mf_{CO_2} = \frac{m_{CO_2}}{m_m} = \frac{10 \text{ kg}}{23 \text{ kg}} = 0.435$$

5 kg O_2
 8 kg N_2
 10 kg CO_2

(b) To find the mole fractions, we need to determine the mole numbers of each component first,

$$N_{O_2} = \frac{m_{O_2}}{M_{O_2}} = \frac{5 \text{ kg}}{32 \text{ kg/kmol}} = 0.156 \text{ kmol}$$

$$N_{N_2} = \frac{m_{N_2}}{M_{N_2}} = \frac{8 \text{ kg}}{28 \text{ kg/kmol}} = 0.286 \text{ kmol}$$

$$N_{CO_2} = \frac{m_{CO_2}}{M_{CO_2}} = \frac{10 \text{ kg}}{44 \text{ kg/kmol}} = 0.227 \text{ kmol}$$

Thus,

$$N_m = N_{O_2} + N_{N_2} + N_{CO_2} = 0.156 \text{ kmol} + 0.286 \text{ kmol} + 0.227 \text{ kmol} = 0.669 \text{ kmol}$$

and

$$y_{O_2} = \frac{N_{O_2}}{N_m} = \frac{0.156 \text{ kmol}}{0.669 \text{ kmol}} = \mathbf{0.233}$$

$$y_{N_2} = \frac{N_{N_2}}{N_m} = \frac{0.286 \text{ kmol}}{0.669 \text{ kmol}} = \mathbf{0.428}$$

$$y_{CO_2} = \frac{N_{CO_2}}{N_m} = \frac{0.227 \text{ kmol}}{0.669 \text{ kmol}} = \mathbf{0.339}$$

(c) The average molar mass and gas constant of the mixture are determined from their definitions:

$$M_m = \frac{m_m}{N_m} = \frac{23 \text{ kg}}{0.669 \text{ kmol}} = \mathbf{34.4 \text{ kg/kmol}}$$

and

$$R_m = \frac{R_u}{M_m} = \frac{8.314 \text{ kJ/kmol} \cdot \text{K}}{34.4 \text{ kg/kmol}} = \mathbf{0.242 \text{ kJ/kg} \cdot \text{K}}$$

Composition of Gas Mixtures

13-14 Determine the mole fractions of a gas mixture that consists of 75 percent CH_4 and 25 percent CO_2 by mass. Also, determine the gas constant of the mixture.

13-15 A gas mixture consists of 8 kmol of H_2 and 2 kmol of N_2 . Determine the mass of each gas and the apparent gas constant of the mixture. *Answers: 16 kg, 56 kg, 1.155 kJ/kg · K*

13-17 A gas mixture consists of 20 percent O_2 , 30 percent N_2 , and 50 percent CO_2 on mass basis. Determine the volumetric analysis of the mixture and the apparent gas constant.

13-34 A gas mixture at 300 K and 200 kPa consists of 1 kg of CO₂ and 3 kg of CH₄. Determine the partial pressure of each gas and the apparent molar mass of the gas mixture.

Properties The molar masses of CO₂ and CH₄ are 44.0 and 16.0 kg/kmol, respectively (Table A-1)

Analysis The mole numbers of the constituents are

$$m_{\text{CO}_2} = 1 \text{ kg} \longrightarrow N_{\text{CO}_2} = \frac{m_{\text{CO}_2}}{M_{\text{CO}_2}} = \frac{1 \text{ kg}}{44 \text{ kg/kmol}} = 0.0227 \text{ kmol}$$

$$m_{\text{CH}_4} = 3 \text{ kg} \longrightarrow N_{\text{CH}_4} = \frac{m_{\text{CH}_4}}{M_{\text{CH}_4}} = \frac{3 \text{ kg}}{16 \text{ kg/kmol}} = 0.1875 \text{ kmol}$$

$$N_m = N_{\text{CO}_2} + N_{\text{CH}_4} = 0.0227 \text{ kmol} + 0.1875 \text{ kmol} = 0.2102 \text{ kmol}$$

$$y_{\text{CO}_2} = \frac{N_{\text{CO}_2}}{N_m} = \frac{0.0227 \text{ kmol}}{0.2102 \text{ kmol}} = 0.108$$

$$y_{\text{CH}_4} = \frac{N_{\text{CH}_4}}{N_m} = \frac{0.1875 \text{ kmol}}{0.2102 \text{ kmol}} = 0.892$$

1 kg CO₂
3 kg CH₄

300 K
200 kPa

Then the partial pressures become

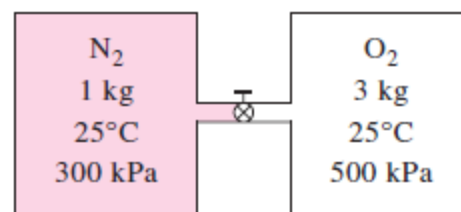
$$P_{\text{CO}_2} = y_{\text{CO}_2} P_m = (0.108)(200 \text{ kPa}) = \mathbf{21.6 \text{ kPa}}$$

$$P_{\text{CH}_4} = y_{\text{CH}_4} P_m = (0.892)(200 \text{ kPa}) = \mathbf{178.4 \text{ kPa}}$$

The apparent molar mass of the mixture is

$$M_m = \frac{m_m}{N_m} = \frac{4 \text{ kg}}{0.2102 \text{ kmol}} = \mathbf{19.03 \text{ kg/kmol}}$$

13–38 A rigid tank that contains 1 kg of N_2 at 25°C and 300 kPa is connected to another rigid tank that contains 3 kg of O_2 at 25°C and 500 kPa. The valve connecting the two tanks is opened, and the two gases are allowed to mix. If the final mixture temperature is 25°C , determine the volume of each tank and the final mixture pressure. *Answers:* 0.295 m^3 , 0.465 m^3 , 422 kPa

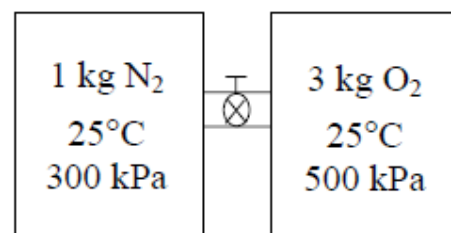


Properties The molar masses of N_2 and O_2 are 28.0 and 32.0 kg/kmol, respectively (Table A-1)

Analysis The volumes of the tanks are

$$V_{N_2} = \left(\frac{mRT}{P} \right)_{N_2} = \frac{(1 \text{ kg})(0.2968 \text{ kPa} \cdot \text{m}^3/\text{kg} \cdot \text{K})(298 \text{ K})}{300 \text{ kPa}} = 0.295 \text{ m}^3$$

$$V_{O_2} = \left(\frac{mRT}{P} \right)_{O_2} = \frac{(3 \text{ kg})(0.2598 \text{ kPa} \cdot \text{m}^3/\text{kg} \cdot \text{K})(298 \text{ K})}{500 \text{ kPa}} = 0.465 \text{ m}^3$$



$$V_{\text{total}} = V_{N_2} + V_{O_2} = 0.295 \text{ m}^3 + 0.465 \text{ m}^3 = 0.76 \text{ m}^3$$

Also,

$$m_{N_2} = 1 \text{ kg} \longrightarrow N_{N_2} = \frac{m_{N_2}}{M_{N_2}} = \frac{1 \text{ kg}}{28 \text{ kg/kmol}} = 0.03571 \text{ kmol}$$

$$m_{O_2} = 3 \text{ kg} \longrightarrow N_{O_2} = \frac{m_{O_2}}{M_{O_2}} = \frac{3 \text{ kg}}{32 \text{ kg/kmol}} = 0.09375 \text{ kmol}$$

$$N_m = N_{N_2} + N_{O_2} = 0.03571 \text{ kmol} + 0.09375 \text{ kmol} = 0.1295 \text{ kmol}$$

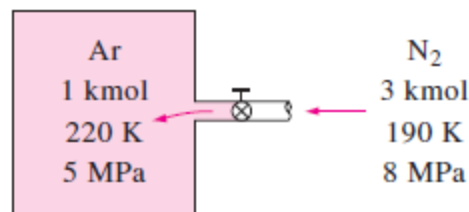
Thus,

$$P_m = \left(\frac{NR_u T}{V} \right)_m = \frac{(0.1295 \text{ kmol})(8.314 \text{ kPa} \cdot \text{m}^3/\text{kmol} \cdot \text{K})(298 \text{ K})}{0.76 \text{ m}^3} = 422.2 \text{ kPa}$$

13-40



A rigid tank contains 1 kmol of Ar gas at 220 K and 5 MPa. A valve is now opened, and 3 kmol of N_2 gas is allowed to enter the tank at 190 K and 8 MPa. The final mixture temperature is 200 K. Determine the pressure of the mixture, using (a) the ideal-gas equation of state and (b) the compressibility chart and Dalton's law.

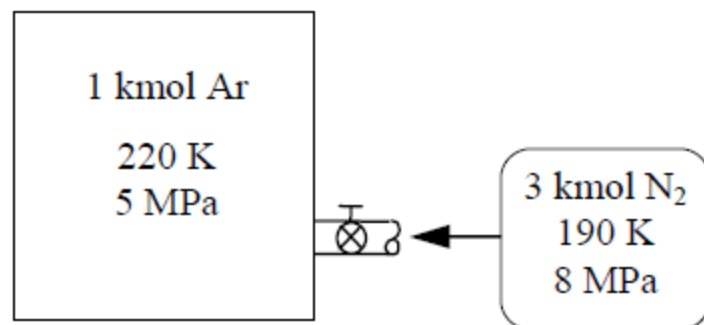


Analysis (a) Under specified conditions both Ar and N_2 will considerably deviate from the ideal gas behavior. Treating the mixture as an ideal gas,

$$\left. \begin{array}{l} \text{Initial state: } P_1 V_1 = N_1 R_u T_1 \\ \text{Final state: } P_2 V_2 = N_2 R_u T_2 \end{array} \right\} P_2 = \frac{N_2 T_2}{N_1 T_1} P_1 = \frac{(4)(200 \text{ K})}{(1)(220 \text{ K})} (5 \text{ MPa}) = \mathbf{18.2 \text{ MPa}}$$

(b) Initially,

$$\left. \begin{array}{l} T_R = \frac{T_1}{T_{cr,Ar}} = \frac{220 \text{ K}}{151.0 \text{ K}} = 1.457 \\ P_R = \frac{P_1}{P_{cr,Ar}} = \frac{5 \text{ MPa}}{4.86 \text{ MPa}} = 1.0278 \end{array} \right\} Z_{Ar} = 0.90 \text{ (Fig. A-15)}$$



Then the volume of the tank is

$$V = \frac{Z N_{Ar} R_u T}{P} = \frac{(0.90)(1 \text{ kmol})(8.314 \text{ kPa} \cdot \text{m}^3 / \text{kmol} \cdot \text{K})(220 \text{ K})}{5000 \text{ kPa}} = 0.33 \text{ m}^3$$



After mixing,

$$\begin{aligned}
 & T_{R,Ar} = \frac{T_m}{T_{cr,Ar}} = \frac{200K}{151.0K} = 1.325 \\
 \text{Ar: } & \left. \begin{aligned}
 \nu_{R,Ar} &= \frac{\nu_{Ar}}{R_u T_{cr,Ar} / P_{cr,Ar}} = \frac{\nu_m / N_{Ar}}{R_u T_{cr,Ar} / P_{cr,Ar}} \\
 &= \frac{(0.33 \text{ m}^3)/(1 \text{ kmol})}{(8.314 \text{ kPa} \cdot \text{m}^3/\text{kmol} \cdot \text{K})(151.0 \text{ K})/(4860 \text{ kPa})} = 1.278
 \end{aligned} \right\} P_R = 0.90 \quad (\text{Fig. A-15}) \\
 \\
 & T_{R,N_2} = \frac{T_m}{T_{cr,N_2}} = \frac{200K}{126.2K} = 1.585 \\
 \text{N}_2: & \left. \begin{aligned}
 \nu_{R,N_2} &= \frac{\nu_{N_2}}{R_u T_{cr,N_2} / P_{cr,N_2}} = \frac{\nu_m / N_{N_2}}{R_u T_{cr,N_2} / P_{cr,N_2}} \\
 &= \frac{(0.33 \text{ m}^3)/(3 \text{ kmol})}{(8.314 \text{ kPa} \cdot \text{m}^3/\text{kmol} \cdot \text{K})(126.2 \text{ K})/(3390 \text{ kPa})} = 0.355
 \end{aligned} \right\} P_R = 3.75 \quad (\text{Fig. A-15})
 \end{aligned}$$

Thus,

$$P_{Ar} = (P_R P_{cr})_{Ar} = (0.90)(4.86 \text{ MPa}) = 4.37 \text{ MPa}$$

$$P_{N_2} = (P_R P_{cr})_{N_2} = (3.75)(3.39 \text{ MPa}) = 12.7 \text{ MPa}$$

and

$$P_m = P_{Ar} + P_{N_2} = 4.37 \text{ MPa} + 12.7 \text{ MPa} = \mathbf{17.1 \text{ MPa}}$$

Properties of Gas Mixtures

13–50 A mixture that is 15 percent carbon dioxide, 5 percent carbon monoxide, 10 percent oxygen, and 70 percent nitrogen by volume undergoes an adiabatic compression process having a compression ratio of 8:1. If the initial state of the mixture is 300 K and 100 kPa, determine the makeup of the mixture on a mass basis and the internal energy change per unit mass of mixture.

Properties 1 The molar masses of CO_2 , CO , O_2 , and N_2 are 44.0, 28.0, 32.0, and 28.0 kg/kmol, respectively (Table A-1). **2** The process is reversible.

Analysis Noting that volume fractions are equal to mole fractions in ideal gas mixtures, the molar mass of the mixture is determined to be

$$\begin{aligned}M_m &= y_{\text{CO}_2} M_{\text{CO}_2} + y_{\text{CO}} M_{\text{CO}} + y_{\text{O}_2} M_{\text{O}_2} + y_{\text{N}_2} M_{\text{N}_2} \\&= (0.15)(44) + (0.05)(28) + (0.10)(32) + (0.70)(28) \\&= 30.80 \text{ kg/kmol}\end{aligned}$$

The mass fractions are

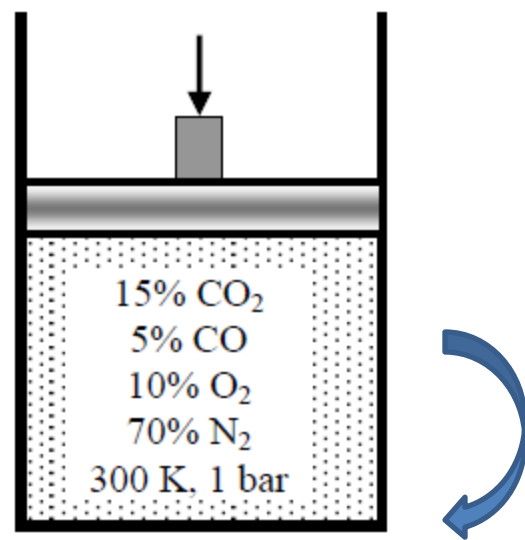
$$\text{mf}_{\text{CO}_2} = y_{\text{CO}_2} \frac{M_{\text{CO}_2}}{M_m} = (0.15) \frac{44 \text{ kg/kmol}}{30.80 \text{ kg/kmol}} = \mathbf{0.2143}$$

$$\text{mf}_{\text{CO}} = y_{\text{CO}} \frac{M_{\text{CO}}}{M_m} = (0.05) \frac{28 \text{ kg/kmol}}{30.80 \text{ kg/kmol}} = \mathbf{0.0454}$$

$$\text{mf}_{\text{O}_2} = y_{\text{O}_2} \frac{M_{\text{O}_2}}{M_m} = (0.10) \frac{32 \text{ kg/kmol}}{30.80 \text{ kg/kmol}} = \mathbf{0.1039}$$

$$\text{mf}_{\text{N}_2} = y_{\text{N}_2} \frac{M_{\text{N}_2}}{M_m} = (0.70) \frac{28 \text{ kg/kmol}}{30.80 \text{ kg/kmol}} = \mathbf{0.6364}$$

The final pressure of mixture is expressed from ideal gas relation to be



$$P_2 = P_1 r \frac{T_2}{T_1} = (100 \text{ kPa})(8) \frac{T_2}{300 \text{ K}} = 2.667 T_2 \quad (\text{Eq. 1})$$

since the final temperature is not known. We assume that the process is reversible as well being adiabatic (i.e. isentropic). Using Dalton's law to find partial pressures, the entropies at the initial state are determined from EES to be:

$$T = 300 \text{ K}, P = (0.2143 \times 100) = 21.43 \text{ kPa} \longrightarrow s_{\text{CO}_2,1} = 5.2190 \text{ kJ/kg.K}$$

$$T = 300 \text{ K}, P = (0.04545 \times 100) = 4.55 \text{ kPa} \longrightarrow s_{\text{CO},1} = 79483 \text{ kJ/kg.K}$$

$$T = 300 \text{ K}, P = (0.1039 \times 100) = 10.39 \text{ kPa} \longrightarrow s_{\text{N}_2,1} = 6.9485 \text{ kJ/kg.K}$$

$$T = 300 \text{ K}, P = (0.6364 \times 100) = 63.64 \text{ kPa} \longrightarrow s_{\text{O}_2,1} = 7.0115 \text{ kJ/kg.K}$$

The final state entropies cannot be determined at this point since the final pressure and temperature are not known. However, for an isentropic process, the entropy change is zero and the final temperature and the final pressure may be determined from

$$\Delta s_{\text{total}} = m f_{\text{CO}_2} \Delta s_{\text{CO}_2} + m f_{\text{CO}} \Delta s_{\text{CO}} + m f_{\text{O}_2} \Delta s_{\text{O}_2} + m f_{\text{N}_2} \Delta s_{\text{N}_2} = 0$$

and using Eq. (1). The solution may be obtained using EES to be $T_2 = 631.4 \text{ K}$, $P_2 = 1684 \text{ kPa}$

The initial and final internal energies are (from EES)



$$T_1 = 300 \text{ K} \longrightarrow \begin{aligned} u_{\text{CO}_2,1} &= -8997 \text{ kJ/kg} \\ u_{\text{CC},1} &= -4033 \text{ kJ/kg} \\ u_{\text{O}_2,1} &= -76.24 \text{ kJ/kg} \\ u_{\text{N}_2,1} &= -87.11 \text{ kJ/kg}, \end{aligned}$$

$$T_2 = 631.4 \text{ K} \longrightarrow \begin{aligned} u_{\text{CO}_2,2} &= -8734 \text{ kJ/kg} \\ u_{\text{CO},2} &= -3780 \text{ kJ/kg} \\ u_{\text{O}_2,2} &= 156.8 \text{ kJ/kg} \\ u_{\text{N}_2,2} &= 163.9 \text{ kJ/kg} \end{aligned}$$

The internal energy change per unit mass of mixture is determined from

$$\begin{aligned} \Delta u_{\text{mixture}} &= \text{mf}_{\text{CO}_2} (u_{\text{CO}_2,2} - u_{\text{CO}_2,1}) + \text{mf}_{\text{CO}} (u_{\text{CO},2} - u_{\text{CO},1}) + \text{mf}_{\text{O}_2} (u_{\text{O}_2,2} - u_{\text{O}_2,1}) + \text{mf}_{\text{N}_2} (u_{\text{N}_2,2} - u_{\text{N}_2,1}) \\ &= 0.2143[(-8734) - (-8997)] + 0.0454[(-3780) - (-4033)] \\ &\quad + 0.1039[156.8 - (-76.24)] + 0.6364[163.9 - (-87.11)] \\ &= \mathbf{251.8 \text{ kJ/kg}} \end{aligned}$$

13-51 Propane and air are supplied to an internal combustion engine such that the air-fuel ratio is 16:1 when the pressure is 95 kPa and the temperature is 30°C. The compression ratio of the engine is 9.5:1. If the compression process is isentropic, determine the required work input for this compression process, in kJ/kg of mixture.

Properties The molar masses of C_3H_8 and air are 44.0 and 28.97 kg/kmol, respectively (Table A-1).

Analysis Given the air-fuel ratio, the mass fractions are determined to be

$$mf_{air} = \frac{AF}{AF+1} = \frac{16}{17} = 0.9412$$

$$mf_{C_3H_8} = \frac{1}{AF+1} = \frac{1}{17} = 0.05882$$

The molar mass of the mixture is determined to be

$$M_m = \frac{1}{\frac{mf_{air}}{M_{air}} + \frac{mf_{C_3H_8}}{M_{C_3H_8}}} = \frac{1}{\frac{0.9412}{28.97 \text{ kg/kmol}} + \frac{0.05882}{44.0 \text{ kg/kmol}}} = 29.56 \text{ kg/kmol}$$

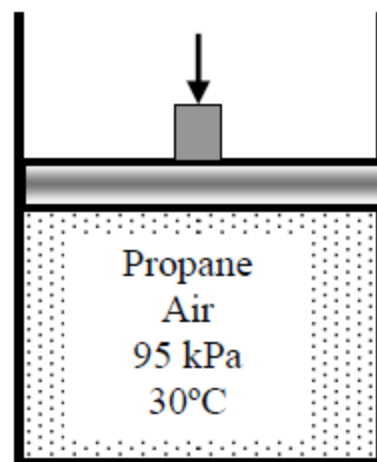
The mole fractions are

$$y_{air} = mf_{air} \frac{M_m}{M_{air}} = (0.9412) \frac{29.56 \text{ kg/kmol}}{28.97 \text{ kg/kmol}} = 0.9606$$

$$y_{C_3H_8} = mf_{C_3H_8} \frac{M_m}{M_{C_3H_8}} = (0.05882) \frac{29.56 \text{ kg/kmol}}{44.0 \text{ kg/kmol}} = 0.03944$$

The final pressure is expressed from ideal gas relation to be

$$P_2 = P_1 r \frac{T_2}{T_1} = (95 \text{ kPa})(9.5) \frac{T_2}{(30 + 273.15) \text{ K}} = 2.977 T_2 \quad (1)$$



since the final temperature is not known. Using Dalton's law to find partial pressures, the entropies at the initial state are determined from EES to be:

$$T = 30^{\circ}\text{C}, P = (0.9606 \times 95) = 91.26 \text{ kPa} \longrightarrow s_{\text{air},1} = 5.7417 \text{ kJ/kg.K}$$

$$T = 30^{\circ}\text{C}, P = (0.03944 \times 95) = 3.75 \text{ kPa} \longrightarrow s_{\text{C}_3\text{H}_8,1} = 6.7697 \text{ kJ/kg.K}$$

The final state entropies cannot be determined at this point since the final pressure and temperature are not known. However, for an isentropic process, the entropy change is zero and the final temperature and the final pressure may be determined from

$$\Delta s_{\text{total}} = mf_{\text{air}} \Delta s_{\text{air}} + mf_{\text{C}_3\text{H}_8} \Delta s_{\text{C}_3\text{H}_8} = 0$$

and using Eq. (1). The solution may be obtained using EES to be

$$T_2 = 654.9 \text{ K}, P_2 = 1951 \text{ kPa}$$

The initial and final internal energies are (from EES)

$$T_1 = 30^{\circ}\text{C} \longrightarrow \begin{array}{l} u_{\text{air},1} = 216.5 \text{ kJ/kg} \\ u_{\text{C}_3\text{H}_8,1} = -2404 \text{ kJ/kg} \end{array} \quad T_2 = 654.9 \text{ K} \longrightarrow \begin{array}{l} u_{\text{air},2} = 477.1 \text{ kJ/kg} \\ u_{\text{C}_3\text{H}_8,2} = -1607 \text{ kJ/kg} \end{array}$$

Noting that the heat transfer is zero, an energy balance on the system gives

$$q_{\text{in}} + w_{\text{in}} = \Delta u_m \longrightarrow w_{\text{in}} = \Delta u_m$$

$$\text{where } \Delta u_m = mf_{\text{air}} (u_{\text{air},2} - u_{\text{air},1}) + mf_{\text{C}_3\text{H}_8} (u_{\text{C}_3\text{H}_8,2} - u_{\text{C}_3\text{H}_8,1})$$

$$\text{Substituting, } w_{\text{in}} = \Delta u_m = (0.9412)(477.1 - 216.5) + (0.05882)[(-1607) - (-2404)] = \mathbf{292.2 \text{ kJ/kg}}$$

13-52 An insulated rigid tank is divided into two compartments by a partition. One compartment contains 2.5 kmol of CO_2 at 27°C and 200 kPa, and the other compartment contains 7.5 kmol of H_2 gas at 40°C and 400 kPa. Now the partition is removed, and the two gases are allowed to mix. Determine (a) the mixture temperature and (b) the mixture pressure after equilibrium has been established. Assume constant specific heats at room temperature for both gases.

CO_2	H_2
2.5 kmol	7.5 kmol
27°C	40°C
200 kPa	400 kPa

Properties The molar masses and specific heats of CO_2 and H_2 are 44.0 kg/kmol, 2.0 kg/kmol, 0.657 kJ/kg $\cdot^\circ\text{C}$, and 10.183 kJ/kg $\cdot^\circ\text{C}$, respectively. (Tables A-1 and A-2b).

Analysis (a) We take both gases as our system. No heat, work, or mass crosses the system boundary, therefore this is a closed system with $Q = 0$ and $W = 0$. Then the energy balance for this closed system reduces to

$$E_{\text{in}} - E_{\text{out}} = \Delta E_{\text{system}}$$

$$0 = \Delta U = \Delta U_{\text{CO}_2} + \Delta U_{\text{H}_2}$$

$$0 = [mc_v(T_m - T_1)]_{\text{CO}_2} + [mc_v(T_m - T_1)]_{\text{H}_2}$$

Using c_v values at room temperature and noting that $m = NM$, the final temperature of the mixture is determined to be

$$(2.5 \times 44 \text{ kg})(0.657 \text{ kJ/kg}\cdot^\circ\text{C})(T_m - 27^\circ\text{C}) + (7.5 \times 2 \text{ kg})(10.183 \text{ kJ/kg}\cdot^\circ\text{C})(T_m - 40^\circ\text{C}) = 0$$

$$T_m = \mathbf{35.8^\circ\text{C}} \quad (308.8 \text{ K})$$

CO_2	H_2
2.5 kmol	7.5 kmol
200 kPa	400 kPa
27°C	40°C



(b) The volume of each tank is determined from

$$V_{\text{CO}_2} = \left(\frac{NR_u T_1}{P_1} \right)_{\text{CO}_2} = \frac{(2.5 \text{ kmol})(8.314 \text{ kPa} \cdot \text{m}^3/\text{kmol} \cdot \text{K})(300 \text{ K})}{200 \text{ kPa}} = 31.18 \text{ m}^3$$

$$V_{\text{H}_2} = \left(\frac{NR_u T_1}{P_1} \right)_{\text{H}_2} = \frac{(7.5 \text{ kmol})(8.314 \text{ kPa} \cdot \text{m}^3/\text{kmol} \cdot \text{K})(313 \text{ K})}{400 \text{ kPa}} = 48.79 \text{ m}^3$$

Thus,

$$V_m = V_{\text{CO}_2} + V_{\text{H}_2} = 31.18 \text{ m}^3 + 48.79 \text{ m}^3 = 79.97 \text{ m}^3$$

$$N_m = N_{\text{CO}_2} + N_{\text{H}_2} = 2.5 \text{ kmol} + 7.5 \text{ kmol} = 10.0 \text{ kmol}$$

and

$$P_m = \frac{N_m R_u T_m}{V_m} = \frac{(10.0 \text{ kmol})(8.314 \text{ kPa} \cdot \text{m}^3/\text{kmol} \cdot \text{K})(308.8 \text{ K})}{79.97 \text{ m}^3} = \mathbf{321 \text{ kPa}}$$



Ethane (C_2H_6) at 20°C and 200 kPa and methane (CH_4) at 45°C and 200 kPa enter an adiabatic mixing chamber. The mass flow rate of ethane is 9 kg/s, which is twice the mass flow rate of methane. Determine (a) the mixture temperature and (b) the rate of entropy generation during this process, in kW/K. Take $T_0 = 25^\circ\text{C}$.

Properties The specific heats of C_2H_6 and CH_4 are 1.7662 kJ/kg $\cdot^\circ\text{C}$ and 2.2537 kJ/kg $\cdot^\circ\text{C}$, respectively. (Table A-2b).

Analysis (a) The enthalpy of ideal gases is independent of pressure, and thus the two gases can be treated independently even after mixing. Noting that $\dot{W} = \dot{Q} = 0$, the steady-flow energy balance equation reduces to

$$\dot{E}_{\text{in}} - \dot{E}_{\text{out}} = \Delta \dot{E}_{\text{system}} \overset{\text{no}}{=} 0 \text{ (steady)} = 0$$

$$\dot{E}_{\text{in}} = \dot{E}_{\text{out}}$$

$$\sum \dot{m}_i h_i = \sum \dot{m}_e h_e$$

$$0 = \sum \dot{m}_e h_e - \sum \dot{m}_i h_i = \dot{m}_{\text{C}_2\text{H}_6} (h_e - h_i)_{\text{C}_2\text{H}_6} + \dot{m}_{\text{CH}_4} (h_e - h_i)_{\text{CH}_4}$$

$$0 = [\dot{m} c_p (T_e - T_i)]_{\text{C}_2\text{H}_6} + [\dot{m} c_p (T_e - T_i)]_{\text{CH}_4}$$

Using c_p values at room temperature and substituting, the exit temperature of the mixture becomes

$$0 = (9 \text{ kg/s})(1.7662 \text{ kJ/kg}\cdot^\circ\text{C})(T_m - 20^\circ\text{C}) + (4.5 \text{ kg/s})(2.2537 \text{ kJ/kg}\cdot^\circ\text{C})(T_m - 45^\circ\text{C})$$

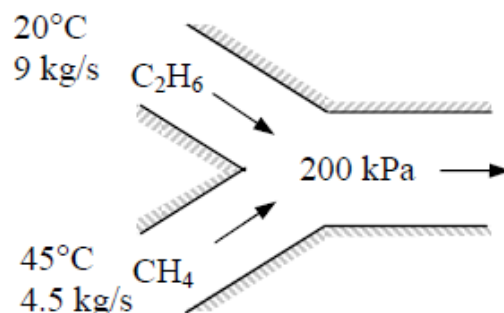
$$T_m = 29.7^\circ\text{C} \quad (302.7 \text{ K})$$

(b) The rate of entropy change associated with this process is determined from an entropy balance on the mixing chamber,

$$\dot{S}_{\text{in}} - \dot{S}_{\text{out}} + \dot{S}_{\text{gen}} = \Delta \dot{S}_{\text{system}} \overset{\text{no}}{=} 0$$

$$[\dot{m}(s_1 - s_2)]_{\text{C}_2\text{H}_6} + [\dot{m}(s_1 - s_2)]_{\text{CH}_4} + \dot{S}_{\text{gen}} = 0$$

$$\dot{S}_{\text{gen}} = [\dot{m}(s_2 - s_1)]_{\text{C}_2\text{H}_6} + [\dot{m}(s_2 - s_1)]_{\text{CH}_4}$$



The molar flow rate of the two gases in the mixture is

$$\dot{N}_{\text{C}_2\text{H}_6} = \left(\frac{\dot{m}}{M} \right)_{\text{C}_2\text{H}_6} = \frac{9 \text{ kg/s}}{30 \text{ kg/kmol}} = 0.3 \text{ kmol/s}$$

$$\dot{N}_{\text{CH}_4} = \left(\frac{\dot{m}}{M} \right)_{\text{CH}_4} = \frac{4.5 \text{ kg/s}}{16 \text{ kg/kmol}} = 0.2813 \text{ kmol/s}$$

Then the mole fraction of each gas becomes

$$y_{\text{C}_2\text{H}_6} = \frac{0.3}{0.3 + 0.2813} = 0.516$$

$$y_{\text{CH}_4} = \frac{0.2813}{0.3 + 0.2813} = 0.484$$

Thus,

$$\begin{aligned} (s_2 - s_1)_{\text{C}_2\text{H}_6} &= \left(c_p \ln \frac{T_2}{T_1} - R \ln \frac{y P_{m,2}}{P_1} \right)_{\text{C}_2\text{H}_6} = \left(c_p \ln \frac{T_2}{T_1} - R \ln y \right)_{\text{C}_2\text{H}_6} \\ &= (1.7662 \text{ kJ/kg} \cdot \text{K}) \ln \frac{302.7 \text{ K}}{293 \text{ K}} - (0.2765 \text{ kJ/kg} \cdot \text{K}) \ln(0.516) = 0.240 \text{ kJ/kg} \cdot \text{K} \end{aligned}$$

$$\begin{aligned} (s_2 - s_1)_{\text{CH}_4} &= \left(c_p \ln \frac{T_2}{T_1} - R \ln \frac{y P_{m,2}}{P_1} \right)_{\text{CH}_4} = \left(c_p \ln \frac{T_2}{T_1} - R \ln y \right)_{\text{CH}_4} \\ &= (2.2537 \text{ kJ/kg} \cdot \text{K}) \ln \frac{302.7 \text{ K}}{318 \text{ K}} - (0.5182 \text{ kJ/kg} \cdot \text{K}) \ln(0.484) = 0.265 \text{ kJ/kg} \cdot \text{K} \end{aligned}$$

Noting that $P_{m,2} = P_{i,1} = 200 \text{ kPa}$ and substituting,

$$\dot{S}_{\text{gen}} = (9 \text{ kg/s})(0.240 \text{ kJ/kg} \cdot \text{K}) + (4.5 \text{ kg/s})(0.265 \text{ kJ/kg} \cdot \text{K}) = \mathbf{3.353 \text{ kW/K}}$$

13–60 A piston–cylinder device contains a mixture of 0.5 kg of H_2 and 1.6 kg of N_2 at 100 kPa and 300 K. Heat is now transferred to the mixture at constant pressure until the volume is doubled. Assuming constant specific heats at the average temperature, determine (a) the heat transfer and (b) the entropy change of the mixture.

Properties The constant pressure specific heats of H_2 and N_2 at 450 K are 14.501 kJ/kg·K and 1.049 kJ/kg·K, respectively. (Table A-2b).

Analysis (a) Noting that $P_2 = P_1$ and $V_2 = 2V_1$,

$$\frac{P_2 V_2}{T_2} = \frac{P_1 V_1}{T_1} \longrightarrow T_2 = \frac{2V_1}{V_1} T_1 = 2T_1 = (2)(300 \text{ K}) = 600 \text{ K}$$

Also $P = \text{constant}$. Then from the closed system energy balance relation,

$$E_{\text{in}} - E_{\text{out}} = \Delta E_{\text{system}}$$

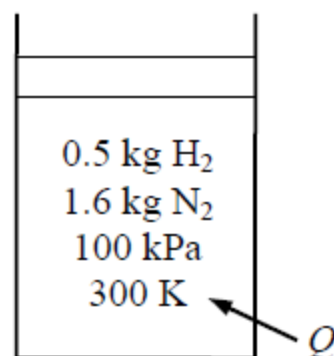
$$Q_{\text{in}} - W_{b,\text{out}} = \Delta U \rightarrow Q_{\text{in}} = \Delta H$$

since W_b and ΔU combine into ΔH for quasi-equilibrium constant pressure processes.

$$Q_{\text{in}} = \Delta H = \Delta H_{H_2} + \Delta H_{N_2} = [mc_{p,\text{avg}}(T_2 - T_1)]_{H_2} + [mc_{p,\text{avg}}(T_2 - T_1)]_{N_2}$$

$$= (0.5 \text{ kg})(14.501 \text{ kJ/kg} \cdot \text{K})(600 - 300) \text{ K} + (1.6 \text{ kg})(1.049 \text{ kJ/kg} \cdot \text{K})(600 - 300) \text{ K}$$

$$= \mathbf{2679 \text{ kJ}}$$



(b) Noting that the total mixture pressure, and thus the partial pressure of each gas, remains constant, the entropy change of the mixture during this process is

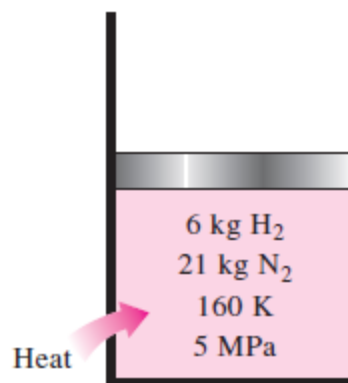
$$\begin{aligned}\Delta S_{\text{H}_2} &= [m(s_2 - s_1)]_{\text{H}_2} = m_{\text{H}_2} \left(c_p \ln \frac{T_2}{T_1} - R \ln \frac{P_2}{P_1} \right)_{\text{H}_2} = m_{\text{H}_2} \left(c_p \ln \frac{T_2}{T_1} \right)_{\text{H}_2} \\ &= (0.5 \text{ kg})(14.501 \text{ kJ/kg} \cdot \text{K}) \ln \frac{600 \text{ K}}{300 \text{ K}} \\ &= 5.026 \text{ kJ/K}\end{aligned}$$

$$\begin{aligned}\Delta S_{\text{N}_2} &= [m(s_2 - s_1)]_{\text{N}_2} = m_{\text{N}_2} \left(c_p \ln \frac{T_2}{T_1} - R \ln \frac{P_2}{P_1} \right)_{\text{N}_2} = m_{\text{N}_2} \left(c_p \ln \frac{T_2}{T_1} \right)_{\text{N}_2} \\ &= (1.6 \text{ kg})(1.049 \text{ kJ/kg} \cdot \text{K}) \ln \frac{600 \text{ K}}{300 \text{ K}} \\ &= 1.163 \text{ kJ/K}\end{aligned}$$

$$\Delta S_{\text{total}} = \Delta S_{\text{H}_2} + \Delta S_{\text{N}_2} = 5.026 \text{ kJ/K} + 1.163 \text{ kJ/K} = \mathbf{6.19 \text{ kJ/K}}$$

13-63 A piston–cylinder device contains 6 kg of H_2 and 21 kg of N_2 at 160 K and 5 MPa. Heat is now transferred to the device, and the mixture expands at constant pressure until the temperature rises to 200 K. Determine the heat transfer during this process by treating the mixture (a) as an ideal gas and (b) as a nonideal gas and using Amagat's law.

Answers: (a) 4273 kJ, (b) 4745 kJ



Properties The molar masses of H_2 and N_2 are 2.0, and 28.0 kg/kmol. (Table A-1).

Analysis From the energy balance relation,

$$E_{in} - E_{out} = \Delta E$$

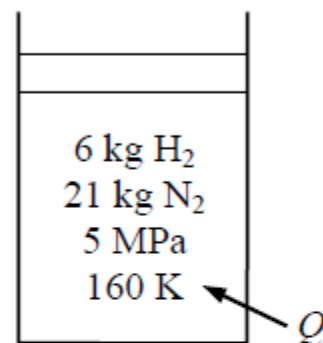
$$Q_{in} - W_{b,out} = \Delta U$$

$$Q_{in} = \Delta H = \Delta H_{H_2} + \Delta H_{N_2} = N_{H_2} (\bar{h}_2 - \bar{h}_1)_{H_2} + N_{N_2} (\bar{h}_2 - \bar{h}_1)_{N_2}$$

since W_b and ΔU combine into ΔH for quasi-equilibrium constant pressure processes

$$N_{H_2} = \frac{m_{H_2}}{M_{H_2}} = \frac{6 \text{ kg}}{2 \text{ kg/kmol}} = 3 \text{ kmol}$$

$$N_{N_2} = \frac{m_{N_2}}{M_{N_2}} = \frac{21 \text{ kg}}{28 \text{ kg/kmol}} = 0.75 \text{ kmol}$$



(a) Assuming ideal gas behavior, the inlet and exit enthalpies of H_2 and N_2 are determined from the ideal gas tables to be

$$H_2: \quad \bar{h}_1 = \bar{h}_{@160 \text{ K}} = 4,535.4 \text{ kJ/kmol}, \quad \bar{h}_2 = \bar{h}_{@200 \text{ K}} = 5,669.2 \text{ kJ/kmol}$$

$$N_2: \quad \bar{h}_1 = \bar{h}_{@160 \text{ K}} = 4,648 \text{ kJ/kmol}, \quad \bar{h}_2 = \bar{h}_{@200 \text{ K}} = 5,810 \text{ kJ/kmol}$$

Thus, $Q_{ideal} = 3 \times (5,669.2 - 4,535.4) + 0.75 \times (5,810 - 4,648) = 4273 \text{ kJ}$

(b) Using Amagat's law and the generalized enthalpy departure chart, the enthalpy change of each gas is determined to be

$$\text{H}_2: \left. \begin{aligned} T_{R_1, \text{H}_2} &= \frac{T_{m,1}}{T_{\text{cr}, \text{H}_2}} = \frac{160}{33.3} = 4.805 \\ P_{R_1, \text{H}_2} &= P_{R_2, \text{H}_2} = \frac{P_m}{P_{\text{cr}, \text{H}_2}} = \frac{5}{1.30} = 3.846 \\ T_{R_2, \text{H}_2} &= \frac{T_{m,2}}{T_{\text{cr}, \text{H}_2}} = \frac{200}{33.3} = 6.006 \end{aligned} \right\} \begin{aligned} Z_{h_1} &\cong 0 \\ Z_{h_2} &\cong 0 \end{aligned} \quad (\text{Fig. A-29})$$

Thus H_2 can be treated as an ideal gas during this process.

$$\text{N}_2: \left. \begin{aligned} T_{R_1, \text{N}_2} &= \frac{T_{m,1}}{T_{\text{cr}, \text{N}_2}} = \frac{160}{126.2} = 1.27 \\ P_{R_1, \text{N}_2} &= P_{R_2, \text{N}_2} = \frac{P_m}{P_{\text{cr}, \text{N}_2}} = \frac{5}{3.39} = 1.47 \\ T_{R_2, \text{N}_2} &= \frac{T_{m,2}}{T_{\text{cr}, \text{N}_2}} = \frac{200}{126.2} = 1.58 \end{aligned} \right\} \begin{aligned} Z_{h_1} &= 1.3 \\ Z_{h_2} &= 0.7 \end{aligned} \quad (\text{Fig. A-29})$$

Therefore,

$$(\bar{h}_2 - \bar{h}_1)_{\text{H}_2} = (\bar{h}_2 - \bar{h}_1)_{\text{H}_2, \text{ideal}} = 5,669.2 - 4,535.4 = 1,133.8 \text{ kJ/kmol}$$

$$\begin{aligned} (\bar{h}_2 - \bar{h}_1)_{\text{N}_2} &= R_u T_{\text{cr}} (Z_{h_1} - Z_{h_2}) + (\bar{h}_2 - \bar{h}_1)_{\text{ideal}} \\ &= (8.314 \text{ kPa} \cdot \text{m}^3 / \text{kmol} \cdot \text{K})(126.2 \text{ K})(1.3 - 0.7) + (5,810 - 4,648) \text{ kJ/kmol} = 1,791.5 \text{ kJ/kmol} \end{aligned}$$

$$\text{Substituting, } Q_{\text{in}} = (3 \text{ kmol})(1,133.8 \text{ kJ/kmol}) + (0.75 \text{ kmol})(1,791.5 \text{ kJ/kmol}) = \mathbf{4745 \text{ kJ}}$$

13–64 Determine the total entropy change and exergy destruction associated with the process described in Prob. 13–63 by treating the mixture (a) as an ideal gas and (b) as a nonideal gas and using Amagat's law. Assume constant specific heats at room temperature and take $T_0 = 30^\circ\text{C}$.

Analysis The entropy generated during this process is determined by applying the entropy balance on an *extended system* that includes the piston-cylinder device and its immediate surroundings so that the boundary temperature of the extended system is the environment temperature at all times. It gives

$$S_{\text{in}} - S_{\text{out}} + S_{\text{gen}} = \Delta S_{\text{system}}$$

$$\frac{Q_{\text{in}}}{T_{\text{boundary}}} + S_{\text{gen}} = \Delta S_{\text{water}} \rightarrow S_{\text{gen}} = m(s_2 - s_1) - \frac{Q_{\text{in}}}{T_{\text{surr}}}$$

Then the exergy destroyed during a process can be determined from its definition $X_{\text{destroyed}} = T_0 S_{\text{gen}}$.

(a) Noting that the total mixture pressure, and thus the partial pressure of each gas, remains constant, the entropy change of a component in the mixture during this process is

$$\Delta S_i = m_i \left(c_p \ln \frac{T_2}{T_1} - R \ln \frac{P_2}{P_1} \right)_i = m_i c_{p,i} \ln \frac{T_2}{T_1}$$

Assuming ideal gas behavior and using c_p values at the average temperature, the ΔS of H_2 and N_2 are determined from

$$\Delta S_{\text{H}_2, \text{ideal}} = (6 \text{ kg})(13.60 \text{ kJ/kg} \cdot \text{K}) \ln \frac{200 \text{ K}}{160 \text{ K}} = 18.21 \text{ kJ/K}$$

$$\Delta S_{\text{N}_2, \text{ideal}} = (21 \text{ kg})(1.039 \text{ kJ/kg} \cdot \text{K}) \ln \frac{200 \text{ K}}{160 \text{ K}} = 4.87 \text{ kJ/K}$$

and

$$S_{\text{gen}} = 18.21 \text{ kJ/K} + 4.87 \text{ kJ/K} - \frac{4273 \text{ kJ}}{303 \text{ K}} = 8.98 \text{ kJ/K}$$

$$X_{\text{destroyed}} = T_0 S_{\text{gen}} = (303 \text{ K})(8.98 \text{ kJ/K}) = 2721 \text{ kJ}$$



(b) Using Amagat's law and the generalized entropy departure chart, the entropy change of each gas is determined to be

$$\text{H}_2: \left. \begin{aligned} T_{R_1, \text{H}_2} &= \frac{T_{m,1}}{T_{\text{cr}, \text{H}_2}} = \frac{160}{33.3} = 4.805 \\ P_{R_1, \text{H}_2} &= P_{R_2, \text{H}_2} = \frac{P_m}{P_{\text{cr}, \text{H}_2}} = \frac{5}{1.30} = 3.846 \\ T_{R_2, \text{H}_2} &= \frac{T_{m,2}}{T_{\text{cr}, \text{H}_2}} = \frac{200}{33.3} = 6.006 \end{aligned} \right\} \begin{aligned} Z_{s_1} &\cong 1 \\ Z_{s_2} &\cong 1 \end{aligned} \quad (\text{Table A-30})$$

Thus H_2 can be treated as an ideal gas during this process.

$$\text{N}_2: \left. \begin{aligned} T_{R_1, \text{N}_2} &= \frac{T_{m,1}}{T_{\text{cr}, \text{N}_2}} = \frac{160}{126.2} = 1.268 \\ P_{R_1, \text{N}_2} &= P_{R_2, \text{N}_2} = \frac{P_m}{P_{\text{cr}, \text{N}_2}} = \frac{5}{3.39} = 1.475 \\ T_{R_2, \text{N}_2} &= \frac{T_{m,2}}{T_{\text{cr}, \text{N}_2}} = \frac{200}{126.2} = 1.585 \end{aligned} \right\} \begin{aligned} Z_{s_1} &= 0.8 \\ Z_{s_2} &= 0.4 \end{aligned} \quad (\text{Table A-30})$$

Therefore,

$$\Delta S_{\text{H}_2} = \Delta S_{\text{H}_2, \text{ideal}} = 18.21 \text{ kJ/K}$$

$$\begin{aligned} \Delta S_{\text{N}_2} &= N_{\text{N}_2} R_u (Z_{s_1} - Z_{s_2}) + \Delta S_{\text{N}_2, \text{ideal}} \\ &= (0.75 \text{ kmol})(8.314 \text{ kPa} \cdot \text{m}^3/\text{kmol} \cdot \text{K})(0.8 - 0.4) + (4.87 \text{ kJ/K}) \\ &= 7.37 \text{ kJ/K} \end{aligned}$$

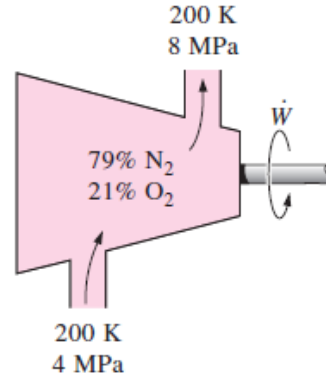
$$\Delta S_{\text{surr}} = \frac{Q_{\text{surr}}}{T_0} = \frac{-4745 \text{ kJ}}{303 \text{ K}} = -15.66 \text{ kJ/K}$$

and

$$S_{\text{gen}} = 18.21 \text{ kJ/K} + 7.37 \text{ kJ/K} - \frac{4745 \text{ kJ}}{303 \text{ K}} = \mathbf{9.92 \text{ kJ/K}}$$

$$X_{\text{destroyed}} = T_0 S_{\text{gen}} = (303 \text{ K})(9.92 \text{ kJ/K}) = \mathbf{3006 \text{ kJ}}$$

13–65 Air, which may be considered as a mixture of 79 percent N_2 and 21 percent O_2 by mole numbers, is compressed isothermally at 200 K from 4 to 8 MPa in a steady-flow device. The compression process is internally reversible, and the mass flow rate of air is 2.9 kg/s. Determine the power input to the compressor and the rate of heat rejection by treating the mixture (a) as an ideal gas and (b) as a nonideal gas and using Amagat's law. *Answers: (a) 115.3 kW, 115.3 kW, (b) 143.6 kW, 94.2 kW*



Properties The molar mass of air is 29.0 kg/kmol. (Table A-1).

Analysis The mass flow rate of air can be expressed in terms of the mole numbers as

$$\dot{N} = \frac{\dot{m}}{M} = \frac{2.90 \text{ kg/s}}{29.0 \text{ kg/kmol}} = 0.10 \text{ kmol/s}$$

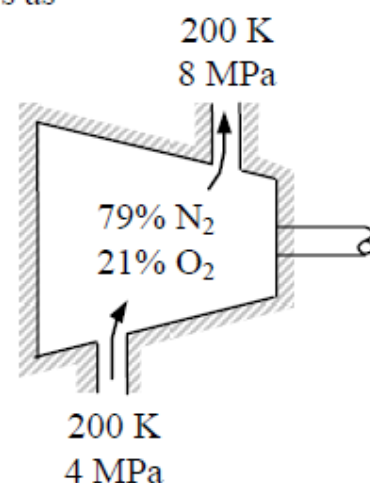
(a) Assuming ideal gas behavior, the Δh and Δs of air during this process is

$$\Delta \bar{h} = 0 \text{ (isothermal process)}$$

$$\begin{aligned} \Delta \bar{s} &= \bar{c}_p \ln \frac{T_2}{T_1} - R_u \ln \frac{P_2}{P_1} = -R_u \ln \frac{P_2}{P_1} \\ &= -(8.314 \text{ kJ/kg} \cdot \text{K}) \ln \frac{8 \text{ MPa}}{4 \text{ MPa}} = -5.763 \text{ kJ/kmol} \cdot \text{K} \end{aligned}$$

Disregarding any changes in kinetic and potential energies, the steady-flow energy balance equation for the isothermal process of an ideal gas reduces to

$$\begin{aligned} \dot{E}_{\text{in}} - \dot{E}_{\text{out}} &= \Delta \dot{E}_{\text{system}}^{\text{no (steady)}} = 0 \\ \dot{E}_{\text{in}} &= \dot{E}_{\text{out}} \\ \dot{W}_{\text{in}} + \dot{N} \bar{h}_1 &= \dot{Q}_{\text{out}} + \dot{N} \bar{h}_2 \\ \dot{W}_{\text{in}} - \dot{Q}_{\text{out}} &= \dot{N} \Delta \bar{h}^{\text{no}} = 0 \longrightarrow \dot{W}_{\text{in}} = \dot{Q}_{\text{out}} \end{aligned}$$



Also for an isothermal, internally reversible process the heat transfer is related to the entropy change by

$$Q = T\Delta S = NT\Delta\bar{s},$$

$$\dot{Q} = \dot{N}T\Delta\bar{s} = (0.10 \text{ kmol/s})(200 \text{ K})(-5.763 \text{ kJ/kmol} \cdot \text{K}) = -115.3 \text{ kW} \rightarrow \dot{Q}_{\text{out}} = 115.3 \text{ kW}$$

Therefore,

$$\dot{W}_{\text{in}} = \dot{Q}_{\text{out}} = \mathbf{115.3 \text{ kW}}$$

(b) Using Amagat's law and the generalized charts, the enthalpy and entropy changes of each gas are determined from

$$\bar{h}_2 - \bar{h}_1 = R_u T_{cr} (Z_{h_1} - Z_{h_2}) + (\bar{h}_2 - \bar{h}_1)_{\text{ideal}}^{\circ}$$

$$\bar{s}_2 - \bar{s}_1 = R_u (Z_{s_1} - Z_{s_2}) + (\bar{s}_2 - \bar{s}_1)_{\text{ideal}}$$

where

$$\text{N}_2: \left. \begin{aligned} P_{R_1} &= \frac{P_{m,1}}{P_{\text{cr},\text{N}_2}} = \frac{4}{3.39} = 1.18 \\ T_{R_1} &= T_{R_2} = \frac{T_m}{T_{\text{cr},\text{N}_2}} = \frac{220}{126.2} = 1.74 \\ P_{R_2} &= \frac{P_{m,2}}{P_{\text{cr},\text{N}_2}} = \frac{8}{3.39} = 2.36 \end{aligned} \right\} \begin{aligned} Z_{h_1} &= 0.4, Z_{s_1} = 0.2 \\ Z_{h_2} &= 0.8, Z_{s_2} = 0.35 \end{aligned} \quad \text{(Tables A-29 and A-30)}$$

$$\text{O}_2: \left. \begin{aligned} P_{R_1} &= \frac{P_{m,1}}{P_{\text{cr},\text{O}_2}} = \frac{4}{5.08} = 0.787 \\ T_{R_1} &= T_{R_2} = \frac{T_m}{T_{\text{cr},\text{O}_2}} = \frac{220}{154.8} = 1.421 \\ P_{R_2} &= \frac{P_{m,2}}{P_{\text{cr},\text{O}_2}} = \frac{8}{5.08} = 1.575 \end{aligned} \right\} \begin{aligned} Z_{h_1} &= 0.4, Z_{s_1} = 0.25 \\ Z_{h_2} &= 1.0, Z_{s_2} = 0.5 \end{aligned} \quad (\text{Tables A-29 and A-30})$$

Then,

$$\begin{aligned} \bar{h}_2 - \bar{h}_1 &= y_i \Delta \bar{h}_i = y_{\text{N}_2} (\bar{h}_2 - \bar{h}_1)_{\text{N}_2} + y_{\text{O}_2} (\bar{h}_2 - \bar{h}_1)_{\text{O}_2} \\ &= (0.79)(8.314)(126.2)(0.4 - 0.8) + (0.21)(8.314)(154.8)(0.4 - 1.0) + 0 \\ &= -494 \text{ kJ/kmol} \end{aligned}$$

$$\begin{aligned} \bar{s}_2 - \bar{s}_1 &= y_i \Delta \bar{s}_i = y_{\text{N}_2} (\bar{s}_2 - \bar{s}_1)_{\text{N}_2} + y_{\text{O}_2} (\bar{s}_2 - \bar{s}_1)_{\text{O}_2} \\ &= (0.79)(8.314)(0.2 - 0.35) + (0.21)(8.314)(0.25 - 0.5) + (-5.763) \\ &= -7.18 \text{ kJ/kmol} \cdot \text{K} \end{aligned}$$

Thus,

$$\dot{Q}_{\text{out}} = -\dot{N}T\Delta\bar{s} = -(0.10 \text{ kmol/s})(200 \text{ K})(-7.18 \text{ kJ/kmol} \cdot \text{K}) = \mathbf{143.6 \text{ kW}}$$

$$\dot{E}_{\text{in}} - \dot{E}_{\text{out}} = \Delta \dot{E}_{\text{system}} \overset{\text{0 (steady)}}{=} 0$$

$$\dot{E}_{\text{in}} = \dot{E}_{\text{out}}$$

$$\dot{W}_{\text{in}} + \dot{N}\bar{h}_1 = \dot{Q}_{\text{out}} + \dot{N}\bar{h}_2$$

$$\dot{W}_{\text{in}} = \dot{Q}_{\text{out}} + \dot{N}(\bar{h}_2 - \bar{h}_1) \longrightarrow \dot{W}_{\text{in}} = 143.6 \text{ kW} + (0.10 \text{ kmol/s})(-494 \text{ kJ/kmol}) = \mathbf{94.2 \text{ kW}}$$

13–67 The combustion of a hydrocarbon fuel with air results in a mixture of products of combustion having the composition on a volume basis as follows: 4.89 percent carbon dioxide, 6.50 percent water vapor, 12.20 percent oxygen, and 76.41 percent nitrogen. Determine the average molar mass of the mixture, the average specific heat at constant pressure of the mixture at 600 K, in $\text{kJ/kmol} \cdot \text{K}$, and the partial pressure of the water vapor in the mixture for a mixture pressure of 200 kPa.

Special Topic: Chemical Potential and the Separation Work of Mixtures

13-73 Brackish water at 12°C with total dissolved solid content of $\text{TDS} = 780 \text{ ppm}$ (a salinity of 0.078 percent on mass basis) is to be used to produce fresh water with negligible salt content at a rate of 280 L/s . Determine the minimum power input required. Also, determine the minimum height to which the brackish water must be pumped if fresh water is to be obtained by reverse osmosis using semipermeable membranes.

13-74 A river is discharging into the ocean at a rate of $400,000 \text{ m}^3/\text{s}$. Determine the amount of power that can be generated if the river water mixes with the ocean water reversibly. Take the salinity of the ocean to be 3.5 percent on mass basis, and assume both the river and the ocean are at 15°C .

13-77 A desalination plant produces fresh water from seawater at 10°C with a salinity of 3.2 percent on mass basis at a rate of $1.4 \text{ m}^3/\text{s}$ while consuming 8.5 MW of power. The salt content of the fresh water is negligible, and the amount of fresh water produced is a small fraction of the seawater used. Determine the second-law efficiency of this plant.

Chapter 13

End of the Problems